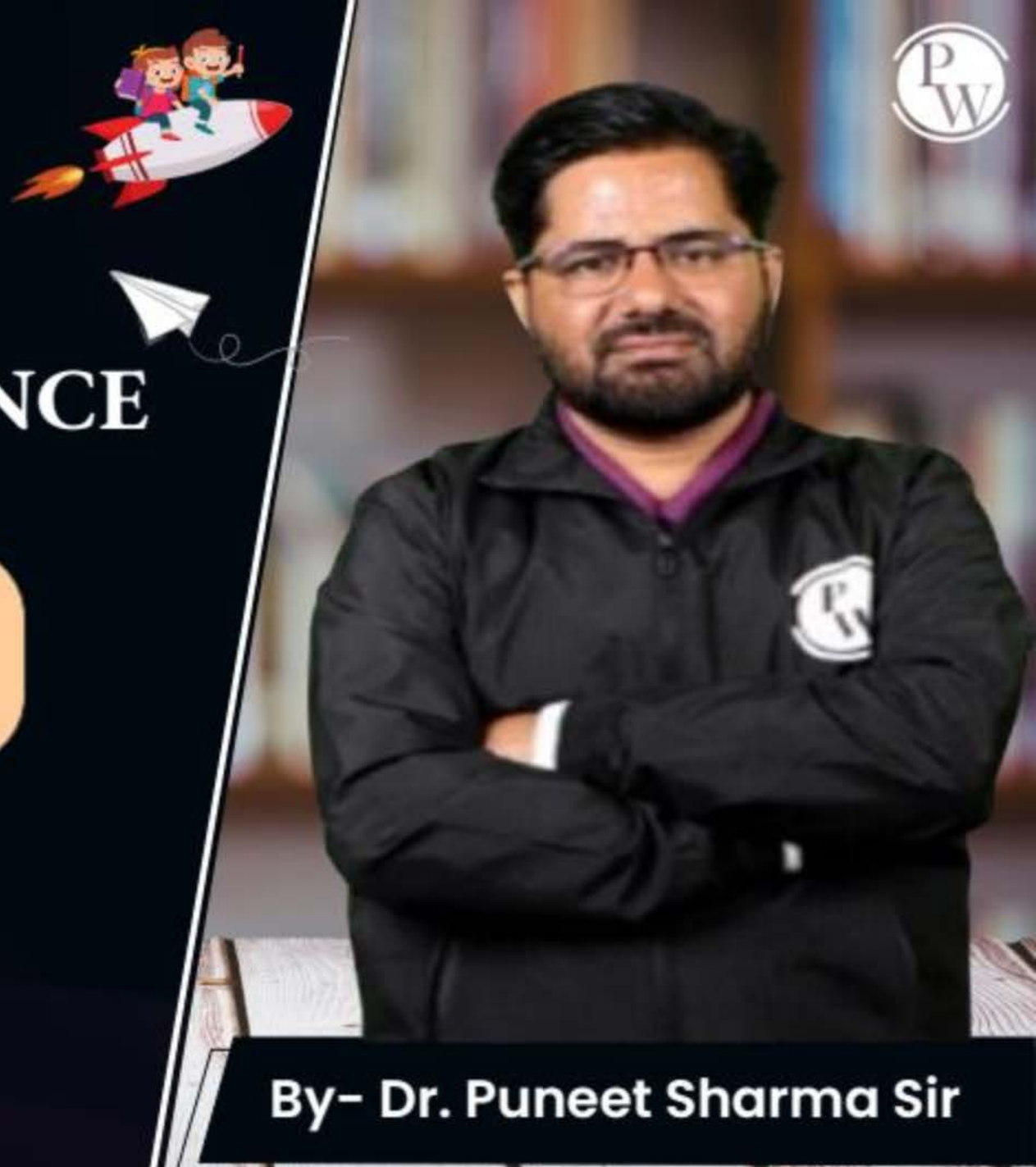
DATA SCIENCE
&
ARTIFICIAL INTELLIGENCE

Probability and Statistics

Statistics - 1

Lecture No. 02



By- Dr. Puneet Sharma Sir

Recap of previous lecture



Topic

1. Tree Based Questions of Probability
2. Quest Based on with & w/o Replacement
3. Hypergeometric Distribution.

Topics to be Covered



Topic BASICS of STATISTICS

Random Variable → whenever we are not sure about the outcome of an experiment, then such types of Experiments are called Random Experiments & Variable associated with R.Exp is called Random Variable

Discrete R.Variable (X) → Counting Related Variables are called D.R.V

for eg No. of Vehicles, No of Students, No of Deaths etc.

Continuous R Variable (x) → if we have infinite possibilities for Random Variable x then it is called C.R.V

for eg Height, Wt, Age etc.

RANDOM VARIABLE

D.R.V (X)

↓

Discrete Prob Dist.

eg (BINOMIAL, POISSON)

↓

Prob. Mass func^n (p.m.f)

$p_i \geq 0$ & $\sum p_i = 1$

C.R.V (x)

↓

Continuous Prob Dist

eg (Normal, uniform, Exponential)

↓

Prob Density func^n (p.d.f)

$f(x) \geq 0$ & $\int_{-\infty}^{\infty} f(x)dx = 1$

p.m.f

① $\text{Mean}(X) = E(X) = \sum p_i x_i$

② $\text{Variance}(X) = E(X^2) - E^2(X)$

③ $SD(\sigma) = +\sqrt{\text{Var}(X)}$

p.d.f

① $\text{Mean}(x) = E(x) = \int_{-\infty}^{\infty} x f(x)\, dx$

② $\text{Var}(x) = E(x^2) - E^2(x)$

③ $SD(\sigma) = +\sqrt{\text{Var}(x)}$

VVV.Sp Note ⇒ $X = \{$ Assume that event as X which is Required $\}$

Prob Distribution → The Table representing Distribution of probabilities at various values of x is called Prob Dist.

eg A coin is tossed thrice then find the prob Dist of Number of Heads ?

$X = \{\text{Number of Heads}\} = \{0, 1, 2, 3\}$

$S = \{\underbrace{(HHH)}_{=3H}, \underbrace{(HHT)}_{2H}, \underbrace{(HTH)}_{2H}, \underbrace{(HTT)}_{1H}, \underbrace{(THH)}_{2H}, \underbrace{(THT)}_{1H}, \underbrace{(TTH)}_{1H}, \underbrace{(TTT)}_{0H}\} \cong 8$ Triplets

$p_1 = P(X=0H) = 1/8$

$p_2 = P(X=1H) = 3/8$

$p_3 = P(X=2H) = 3/8$

$p_4 = P(X=3H) = 1/8$

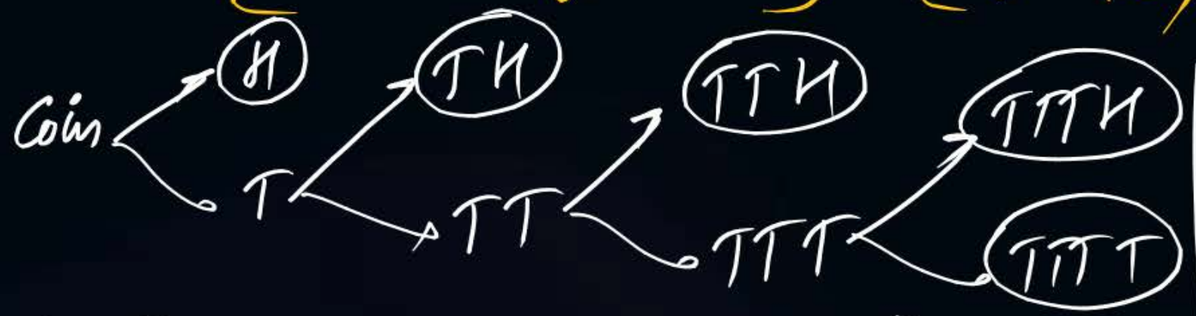
So Prob Dist is

X:	0	1	2	3
$P(X)$:	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

$\because \sum p_i = 1$ So all correct.

Q: A Coin is tossed (until) Head appears or tail appears 4 times in succession then (Find) the Prob Dist. of Number of tosses?

Sol: $X = \{\text{Number of tosses}\} = \{1, 2, 3, 4, 4\} = \{1, 2, 3, 4\}$



$S = \{H, TH, TTH, TTTH, TTTT\}$

$p_1 = P(X = 1 \text{ toss}) = P(H) = \frac{1}{2}$

$p_2 = P(X = 2 \text{ tosses}) = P(TH) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

$p_3 = P(X = 3 \text{ tosses}) = P(TTH) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$

$p_4 = P(X = 4 \text{ tosses})$

$= P[(TTTH) \text{ or } (TTTT)] = \frac{1}{16} + \frac{1}{16} = \frac{1}{8}$

So Req. Prob Dist is

X	1	2	3	4
$P(X)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$

$\because \Sigma p_i = 1$ So all Correct.

Mean (X) / Central Value / Average Value / Expected Value →

$(\bar{X})$ (μ_X) $E(X)$

(M-I) $\bar{X} = \frac{\Sigma X}{N} = \frac{\text{sum of observations}}{\text{No. of observation}}$, (M-II) $E(X) = \frac{\Sigma p_i x_i}{\Sigma p_i} = \Sigma p_i x_i$

i.e. $\bar{X} = E(X) = \frac{\Sigma X}{N}$ or $\Sigma p_i x_i$ i.e. $E(X)$ is the Av of Prob Dist.

Variance (X) → it measures the spread of Distribution about it's Central Value

(MSD) i.e. Variance is Average of Squares of Deviations Calculated from Central Value

$$Var(X) = \frac{\Sigma (X - \bar{X})^2}{N} = \ldots\ldots\ldots = E(X^2) - E^2(X)$$

Standard Deviation (RMSD) → σ → "it has the same physical significance as that of Variance" & it is defined as $\boxed{S.D = +\sqrt{Var(X)}}$

Note: ① $E(X) = \sum p_i x_i$, $E(X^2) = \sum p_i x_i^2$, $E(X^3) = \sum p_i x_i^3$ .....

② Var & SD ≥ 0 (T) (equality holds in case of constant Dist.)

③ Var or S.D $\propto \frac{1}{\text{Consistency}}$ (it is obvious)

Mode :- the Data having highest frequency is called Mode

Median :- After arranging the Data either in Ascending order or in Descending order the Middle most Value is called Median.

Case I: if N = odd then $Md = \left(\frac{N}{2}+1\right)^{th}$ observation

Case II: if N = even then $Md = \frac{\left(\frac{n}{2}\right)^{th} \text{ observation} + \left(\frac{N}{2}+1\right)^{th} \text{ observation}}{2}$

eg 2, 2, 3, 4, 4, 4, 5, 5, 6, 6, 7, 7, 7, 7, 9 ⇒ mode = 7

N = 16 so $Md = \frac{\left(\frac{16}{2}\right)^{th} + \left(\frac{16}{2}+1\right)^{th}}{2} = \frac{8^{th}+9^{th}}{2} = \frac{5+6}{2} = 5.5$

(9) 8 5 6 5

(6) 6 (9) 5 5

eg Consider 4 children having weights 9kg, 13kg, 16kg, 22kg

Then Average weight $(\bar{X}) = \frac{\Sigma X}{N} = \frac{9+13+16+22}{4} = 15$ kg



Now Average of deviations from central Value $= \frac{(9-15)+(13-15)+(16-15)+(22-15)}{4} = 0$ kg

Now Modulus of Av of Deviations from central Value $= \frac{|-6|+|-2|+|1|+|7|}{4} = 4$ kg ✓

again Average of squares of deviations from central $= \frac{(-6)^2+(-2)^2+(1)^2+(7)^2}{4} = 22.5\,(kg)^2$

$=$ Variance (let)

So Standard Deviation $= \sqrt{22.5\,kg^2} = 4.75$ kg ✓✓

Eg: A coin is tossed thrice then find Mean, Variance & SD of no. of Heads

Sol: $X = \{\text{Number of Heads}\} = \{0, 1, 2, 3\}$

X:	0	1	2	3
$P(x)$:	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

$$E(X) = \sum p_i x_i = p_1 x_1 + p_2 x_2 + p_3 x_3 + p_4 x_4$$

$$= \frac{1}{8}(0) + \frac{3}{8}(1) + \frac{3}{8}(2) + \frac{1}{8}(3)$$

$$= \frac{3}{2} = 1.5$$

Average Number of Heads = 1.5

$$E(X^2) = \sum p_i x_i^2$$

$$= p_1 x_1^2 + p_2 x_2^2 + p_3 x_3^2 + p_4 x_4^2$$

$$= \frac{1}{8}(0)^2 + \frac{3}{8}(1)^2 + \frac{3}{8}(2)^2 + \frac{1}{8}(3)^2$$

$$= 3$$

$$\therefore Var(X) = E(X^2) - E^2(X)$$

$$= 3 - \left(\frac{3}{2}\right)^2 = \frac{3}{4} = 0.75$$

$$\& \; SD(\sigma) = +\sqrt{Var(X)} = \frac{\sqrt{3}}{2}$$

WRONG APP

$X = \{0,1,2,3\} \Rightarrow \bar{X} = \frac{\Sigma X}{N} = \frac{0+1+2+3}{4} = 1.5$

$Var(X) = \frac{\Sigma (X-\bar{X})^2}{N} = \frac{(0-1.5)^2+(1-1.5)^2+(2-1.5)^2+(3-1.5)^2}{4} = 1.25 \neq 0.75$

Correct App →

$X = \{0,1,1,1,2,2,2,3\} \Rightarrow \bar{X} = \frac{0+1+1+1+2+2+2+3}{8} = 1.5$

$Var(X) = \frac{\Sigma (X-\bar{X})^2}{N} = \frac{(0-1.5)^2+3(1-1.5)^2+3(2-1.5)^2+(3-1.5)^2}{8} = 0.75$

⊛ it is advisable to use Mature Method (Prob Dist Method)

Analysis:- When a Coin is tossed thrice, Av No of Heads = 1·5, Var = 0·75, SD = $\frac{\sqrt{3}}{2}$

i.e. " " " " once, Av No of Heads = 0·5, Var = 0·25, SD = $\frac{\sqrt{3}/2}{\sqrt{3}} = 0.5$

$\Rightarrow$ When a Coin is tossed 10000 times, Av No of Heads = 5000, Var = 2500, SD = $0.5\sqrt{10000} = 50$

i.e. $(5000-50) \leq$ Av No. of Heads $\leq (5000+50)$

or $4950 \leq$ Av No. of Heads ≤ 5050

i.e. on an Average, No of Heads lies in b/w 4950 & 5050 in general

Q. Which of the following Batsman is more Consistent?

2014 / 2M

Batsman	AV	SD
K	43.2	5.21
L	65	6.71
M	57	4.23
N	41.5	5.11

$\because$ Consistency $\propto \frac{1}{S.D}$

Ans = M

Q. If the Diff b/n Expected Value of the square of Random Variable & Sq of the Expected Value is given as R then

2018

(a) $R=0$ (b) $R>0$

(c) $R\geq 0$ (d) $R<0$

Let X is Random Variable then ATQ,

$R = E(X^2) - (E(X))^2$

$= Var(X)$

Q8 (PYQ) Marks obtained by 100 students in a test is given in the following Table then find Mean, Median & Mode of Marks obtained ?

Marks obtained (X)	No. of students (N)
25	20
30	20
35	40
40	20
$\Sigma X = ?$	$N = 100$

$25, 25, 25 \ldots\ldots 25$, $30, 30, 30 \ldots\ldots 30$, $35, 35, 35 \ldots\ldots 35$, $40, 40, 40 \ldots\ldots 40$

20 students, 20 students, 40 students, 20 students.

Mode = 35 Marks, $Md = \dfrac{\left(\frac{100}{2}\right)^{th} + \left(\frac{100}{2}+1\right)^{th}}{2} = \dfrac{50^{th} + 51^{st}}{2} = \dfrac{35+35}{2} = 35$ Marks

Av. Marks $\bar{X} = \dfrac{\Sigma X}{N} = \dfrac{(25+25+\ldots+25)+(30+30+\ldots+30)+(35+35+\ldots+35)+(40+40+\ldots 40)}{100}$

$= \dfrac{20\times25 + 20\times30 + 40\times35 + 20\times40}{100} = \dfrac{3300}{100} = 33$

THANK - YOU